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> #Date: November 19, 2024
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> #This code reads in the Car Mileage Data Set. Additionally,  
it
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```
> #finds the 98% confidence interval for the average
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```
> #miles per gallon for all cars in the population of cars
```

```
> #manufactured regardless of country. It uses the z-test
```

```
> #because sample size is 56, which is greater than 30.
```

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>
```

```
> #Reading in Car Mileage data set, creating data frame
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```
> mpg<-read.csv("Car Mileage.csv")
```

```
> mpg
```

```
Country MPG
```

```
1 United States 18
```

```
2 United States 15
```

```
3 United States 18
```

```
4 United States 16
```

```
5 United States 17
```

```
6 United States 15
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7 United States 14
```

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8 United States 14
```

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9 United States 14
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10 United States 15
```

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11 United States 15
```

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12 United States 14
```

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13 United States 15
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14 United States 14
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```
15 United States 22
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16 United States 18
17 United States 21
18 United States 21
19 United States 10
20 United States 10
21 United States 11
22 United States 9
23 United States 28
24 United States 25
25 United States 19
26 United States 16
27 United States 17
28 United States 19
29 Japan 24
30 Japan 27
31 Japan 27
32 Japan 25
33 Japan 31
34 Japan 35
35 Japan 24
36 Japan 19
37 Japan 28
38 Japan 23
39 Japan 27
40 Japan 20
41 Japan 22
42 Japan 18
43 Japan 20

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44    Japan 31
45    Japan 32
46    Japan 31
47    Japan 32
48    Japan 24
49    Japan 26
50    Japan 29
51    Japan 24
52    Japan 24
53    Japan 33
54    Japan 33
55    Japan 32
56    Japan 28
```

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>
```

```
> #Finding the 98% confidence interval for the average
> #miles per gallon for all cars in the population of cars
> #manufactured regardless of country
> #Uses z-test function, because sample size is 56 (56>30)
> s_mpg<-sd(na.omit(mpg$MPG))
> CI_98_mean_mpg<-z.test(mpg$MPG, sigma.x=s_mpg,
conf.level=.98)
> CI_98_mean_mpg
```

One-sample z-Test

data: mpg\$MPG

$z = 23.509$, $p\text{-value} < 2.2e-16$

alternative hypothesis: true mean is not equal to 0

98 percent confidence interval:

19.45289 23.72568

sample estimates:

mean of x

21.58929

```
> CI_98_mean_mpg$conf.int
```

```
[1] 19.45289 23.72568
```

```
attr(,"conf.level")
```

```
[1] 0.98
```

```
>
```